

Self-Referencing Piezoresistive Pressure Sensor

Executive Technology Brief

Domain: Piezoresistive Sensing + Tubing Mechanics + Drift Compensation

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Chronic ICP monitoring sensors suffer from drift

Mechanism: Piezoresistive MEMS sensor with self-referencing Wheatstone bridge for drift compensation (R1 fix)

Architecture: Catheter-integrated piezoresistive pressure sensor with self-referencing architecture for drift compensation. R1 fix: original P-27 suffered from chronic drift; R1 redesign adds self-referencing bridge for drift correction.

Governing Model: Piezoresistive pressure sensing with Wheatstone bridge and self-referencing drift compensation. Bridge output proportional to pressure-induced resistance change; dummy element cancels thermal drift.

Key Equations:

$dR/R = \pi_{iL} * \sigma_{iL} + \pi_{iT} * \sigma_{iT}$ [piezoresistive effect]

$V_{out} = (dR / (4*R + 2*dR)) * V_{ex}$ [bridge output]

$V_{out} = ((R_{active} - R_{dummy}) / (4*R)) * V_{ex}$ [self-referencing bridge]

$P_{measured} = V_{out} / (S * V_{ex})$ [pressure from voltage, S = sensitivity]

Kill Condition: Self-referencing cannot reduce drift below 1 mmHg/month (insufficient for clinical utility)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.